Thales' theorem

Parallel lines cutting equal pieces from one transversal cut equal pieces from every other transversal.

LESSON 12 | 1 × 40 MIN | GEOMETRY 8, TEMA 9 | STAGE 9 · 13.2, 13.4

LESSON CLOCK

4' | 13' | 8' | 13' | 2'

Warm-up	4 min
Explanation	13 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- State Thales' theorem precisely.
- Use it to divide a segment into a given number of equal parts.
- Use the proportional form to find an unknown length.
- Recognise when the theorem does *not* apply.

TEXTBOOK

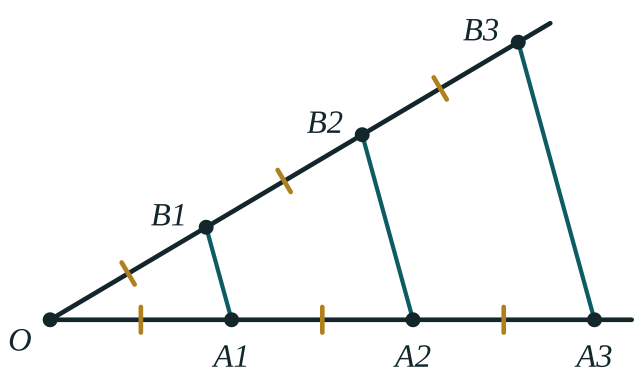
National	Тема 9, pp. 23–25
Cambridge	Learner's Book pp. 278–299
Workbook	Workbook 13.2

01 Explanation

The theorem

THALES' THEOREM

If parallel lines cut equal segments on one transversal, then they cut equal segments on every other transversal.



Three parallel lines cut equal segments on the lower ray, and therefore cut equal segments on the upper ray as well.

In the figure, $OA_1 = A_1A_2 = A_2A_3$ on the lower ray forces $OB_1 = B_1B_2 = B_2B_3$ on the upper one — even though the two rays make different angles and the segments have different lengths.

THE PROPORTIONAL FORM

More generally, parallel lines cut **proportional** segments:

$$\frac{\mathrm{OA}_1}{\mathrm{OA}_2} = \frac{\mathrm{OB}_1}{\mathrm{OB}_2} \quad \text{and} \quad \frac{\mathrm{OA}_1}{\mathrm{A}_1\mathrm{A}_2} = \frac{\mathrm{OB}_1}{\mathrm{B}_1\mathrm{B}_2}$$

This is the form you will use for calculations; the equal-segments version is the special case where the ratio is 1.

Why it is true

Through B_1 draw a line parallel to the lower ray; it cuts the next parallel at a point P. Then OA_1B_1 and B_1PB_2 are triangles with equal corresponding angles (alternate and corresponding angles on the parallels) and one equal side ($A_1A_2 = B_1P$, opposite sides of a parallelogram). They are congruent by ASA, so $OB_1 = B_1B_2$. Repeating the argument along the ray finishes the proof. ∎

Dividing a segment into equal parts

This is the classic construction, and it needs no measurement at all.

1. To divide AB into 5 equal parts, draw any ray from A.
2. With compasses, step off 5 equal segments along the ray, ending at P_5.
3. Join P_5 to B.
4. Through P_1, P_2, P_3, P_4 draw lines parallel to P_5B.
5. By Thales' theorem these lines cut AB into 5 equal parts.

WHY THIS MATTERS

Bisecting a segment is easy with compasses. Cutting it into **five** equal parts is not — and Thales' theorem does it exactly, with no arithmetic and no ruler markings.

When it does not apply

THE LINES MUST BE PARALLEL

> Three lines that merely look evenly spread do not give equal cuts. If the figure does not say “parallel” — or you cannot prove it — the theorem is not available.

And the equal segments must be measured *along the transversals*, not along the parallel lines themselves.

02 Worked examples

EXAMPLE 1 ***Parallel lines cut a transversal so that $OA_1 = A_1A_2 = A_2A_3 = 4$ cm, and on a second transversal $OB_1 = 3$ cm. Find OB_3.***

1. $OB_1 = B_1B_2 = B_2B_3$

   Thales’ theorem — equal cuts on one transversal give equal cuts on the other.

2. $B_1B_2 = B_2B_3 = 3\ cm$

3. $OB_3 = 3 \cdot 3 = 9\ cm$

ANSWER

$OB_3 = 9\ cm$

EXAMPLE 2 ***In the proportional form, $OA_1 = 6$, $OA_2 = 15$ and $OB_1 = 8$. Find OB_2.***

1. $$\frac{OA_1}{OA_2} = \frac{OB_1}{OB_2}$$

   Set up the proportion.

2. $$\frac{6}{15} = \frac{8}{OB_2}$$

   Substitute.

3. $6 \cdot OB_2 = 15 \cdot 8 = 120$

   Cross-multiply.

4. $OB_2 = 20$

ANSWER

$OB_2 = 20$

EXAMPLE 3 **Describe how to divide a segment of length 7 cm into 3 equal parts without measuring.**

1. Draw a ray from one end A at any convenient angle.

   The angle does not matter.

2. Step off three equal compass widths along the ray: P_1, P_2, P_3.

   Any width will do.

3. Join P_3 to the other end B.

4. Through P_1 and P_2 draw lines parallel to P_3B.

   They cut AB into three equal parts by Thales.

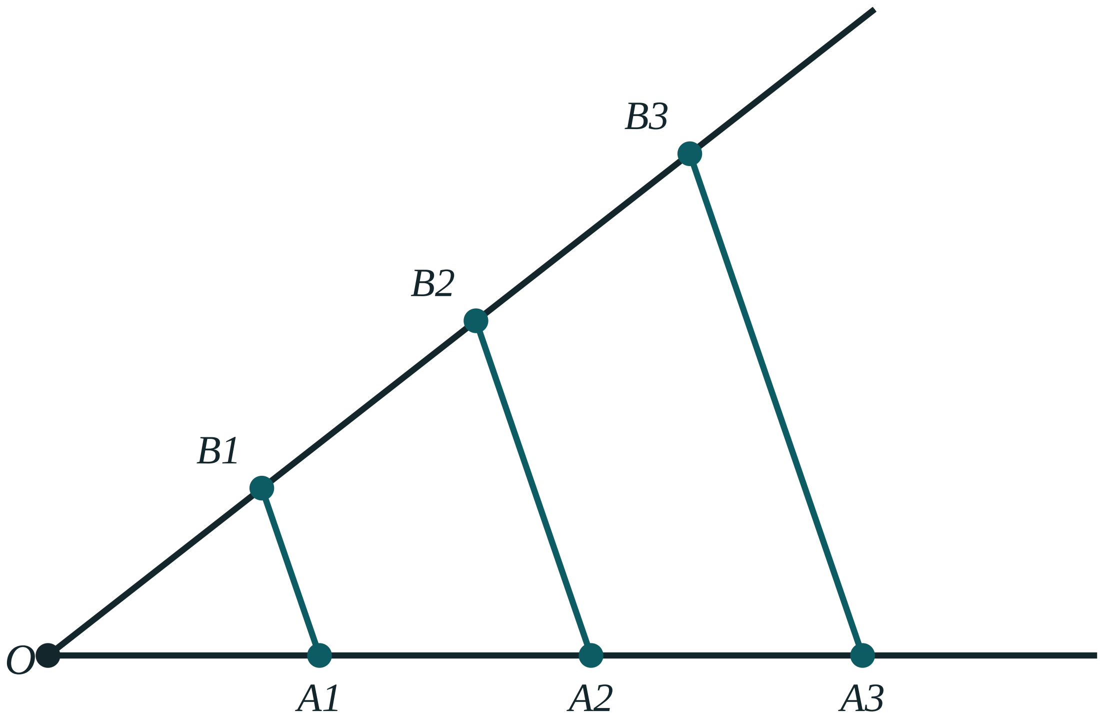
ANSWER

Each part is $\frac{7}{3} \approx 2.33\ cm$ — obtained exactly, without measuring it.

03 Interactive model

Change the angle and the spacing. The ratios in the readout never move.

Thales’ theorem



OA_1 **88** | A_1A_2 **88** | OB_1 **88** | B_1B_2 **88** | $OA_1 : OA_2$ **1 : 2** | $OB_1 : OB_2$ **1 : 2**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Thales’ theorem	Fales teoremasi	Теорема Фалеса
Transversal	Kesuvchi	Секущая
Equal segments	Teng kesmalar	Равные отрезки
Proportional	Proporsional	Пропорциональный
Ratio	Nisbat	Отношение
Divide a segment	Kesmani bo‘lish	Разделить отрезок
Construction	Yasash	Построение
Ray	Nur	Луч

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Thales’ theorem requires the cutting lines to be:

- equal in length
- parallel
- perpendicular to a transversal
- equally spaced by eye

If parallels cut 5 cm, 5 cm, 5 cm on one transversal and 3 cm on the first piece of another, the second piece is:

- *3 cm*
- *5 cm*
- *9 cm*
- not determined

$OA_1 : OA_2 = 2 : 5$ and $OB_1 = 6$. Then OB_2 is:

- 10
- 15
- 12
- 30

Thales’ theorem is most useful for:

- finding areas
- dividing a segment into n equal parts
- proving triangles congruent
- measuring angles

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *Parallels cut 3 cm, 3 cm on one transversal and 4 cm on the first piece of another. Find the second piece.*

   ANSWER *4 cm*

2. $OA_1 = A_1A_2 = 5$ *cm and* $OB_1 = 7$ *cm. Find* OB_2.

   ANSWER *14 cm*

3. *Three parallels cut equal pieces of 6 cm on one line. Find the total for four pieces.*

   ANSWER *24 cm*

4. *Does Thales' theorem apply to three lines that are not parallel?*

   ANSWER No.

5. $OA_1 : OA_2 = 1 : 3$ *and* $OB_1 = 4$. *Find* OB_2.

   ANSWER *12*

6. *Equal cuts of 2 cm are made on one transversal. What are the cuts on another?*

   ANSWER Equal to one another, though not necessarily 2 cm.

7. *How many parallels are needed to cut a segment into 4 equal parts?*

   ANSWER *3* (plus the two ends)

Medium · the standard question

7 PROBLEMS

1. *$OA_1 = A_1A_2 = A_2A_3 = 4$ cm and $OB_1 = 3$ cm. Find OB_3.*

   ANSWER *9 cm*

2. *$OA_1 = 6$, $OA_2 = 15$, $OB_1 = 8$. Find OB_2.*

   ANSWER *20*

3. *$OA_1 = 4$, $A_1A_2 = 6$, $OB_1 = 10$. Find B_1B_2.*

   ANSWER *15*

4. *A segment of 12 cm is divided into 4 equal parts by Thales' construction. Find each part.*

   ANSWER *3 cm*

5. *$OA_1 : A_1A_2 = 3 : 4$ and $B_1B_2 = 12$. Find OB_1.*

   ANSWER *9*

6. *Describe how to divide a 7 cm segment into 3 equal parts without measuring.*

   ANSWER Ray, three equal compass steps, join the last to B, draw parallels.

7. *Parallels cut $OA_1 = 5$, $A_1A_2 = 5$, $A_2A_3 = 5$ and $OB_3 = 21$. Find OB_1.*

   ANSWER *7*

Hard · stretch and reason

7 PROBLEMS

1. *Prove Thales' theorem for two equal segments using a parallelogram and ASA.*

   ANSWER Through B_1 draw a parallel to the first ray; the parallelogram gives $A_1A_2 = B_1P$, then ASA gives $OB_1 = B_1B_2$.

2. *Divide a segment in the ratio 2 : 3 using Thales' construction.*

   ANSWER Step off 5 equal parts on a ray, join the 5th to B, and draw the parallel through the 2nd point.

3. $OA_1 = x, A_1A_2 = x + 2, OB_1 = 6, B_1B_2 = 9$. *Find* x.

   ANSWER $\frac{x}{x+2} = \frac{6}{9} \Longrightarrow 9x = 6x + 12 \Longrightarrow x = 4$

4. *In* $\triangle ABC$ *a line parallel to* BC *cuts* AB *at* M *and* AC *at* N, *with* $AM : MB = 2 : 3$ *and* $AN = 8$. *Find* NC.

   ANSWER $\frac{AN}{NC} = \frac{2}{3} \Longrightarrow NC = 12$

5. *A ladder rests against a wall and three equally spaced horizontal rungs meet it. Explain why they cut the wall into equal parts.*

   ANSWER The rungs are parallel and cut equal pieces on the ladder, so by Thales' theorem they cut equal pieces on the wall.

6. *Show that the construction for n equal parts needs no measurement of AB itself.*

   ANSWER Only equal compass steps on the auxiliary ray are used; Thales' theorem transfers the equality to AB.

7. *Three parallels cut a transversal into pieces 3 cm and 5 cm, and cut another transversal into pieces* p *and* $p + 4$. *Find* p.

   ANSWER $\frac{3}{5} = \frac{p}{p+4} \Longrightarrow 5p = 3p + 12 \Longrightarrow p = 6$

07 Homework

Homework — 5 problems

Geometry 8, Тема 9, pp. 23–25. Question 5 is a construction — bring it drawn.

1. $OA_1 = A_1A_2 = A_2A_3 = 3$ *cm and* $OB_1 = 5$ *cm. Find* OB_3.
2. $OA_1 = 8$, $OA_2 = 20$, $OB_1 = 6$. *Find* OB_2.
3. $OA_1 : A_1A_2 = 2 : 5$ *and* $B_1B_2 = 20$. *Find* OB_1.
4. *In* $\triangle ABC$, $MN \parallel BC$ *with* $AM : MB = 3 : 4$ *and* $AN = 9$. *Find* NC.
5. *Construct the division of a 9 cm segment into 5 equal parts and state the theorem you used.*